

Homework6_12412639_ WangSiyu

1. Algorithm formulas

Standard gradient descent:

$$x^{(k+1)} = x^{(k)} - \alpha \nabla f(x^{(k)})$$

Gradient descent with momentum:

$$x^{(k+1)} = x^{(k)} - \alpha \nabla f(x^{(k)}) + \beta (x^{(k)} - x^{(k-1)})$$

where $\alpha > 0$ is the constant step size and $\beta \in [0, 1)$ is the momentum coefficient. When $\beta = 0$, the method becomes standard gradient descent.

Velocity form:

$$v^{(k+1)} = \beta v^{(k)} - \alpha \nabla f(x^{(k)})$$

$$x^{(k+1)} = x^{(k)} + v^{(k+1)}$$

2. Test functions and gradients

Function 1:

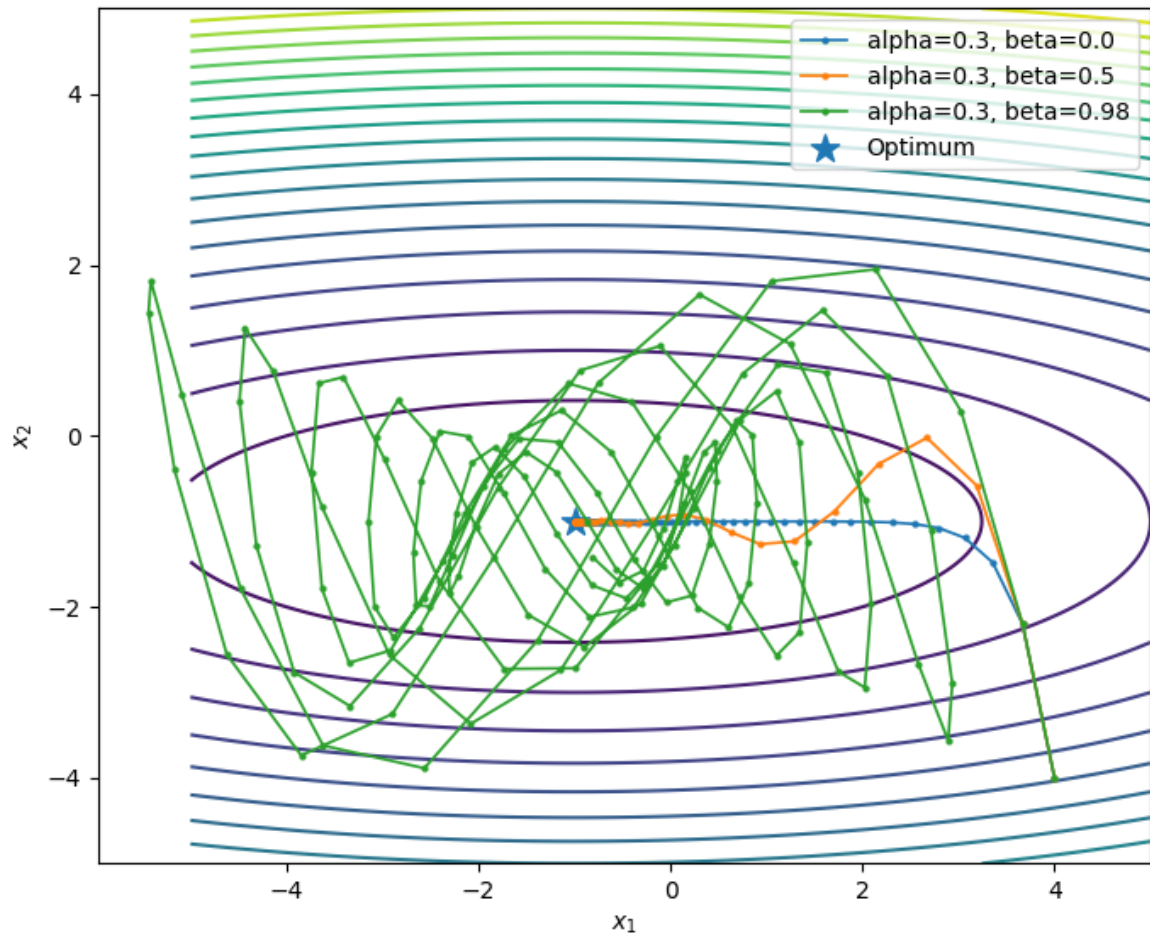
$$f_1(x) = \frac{(x_1 + 1)^2}{9} + (x_2 + 1)^2$$

$$\nabla f_1(x) = \left[\frac{2(x_1+1)}{9} \quad 2(x_2 + 1) \right]$$

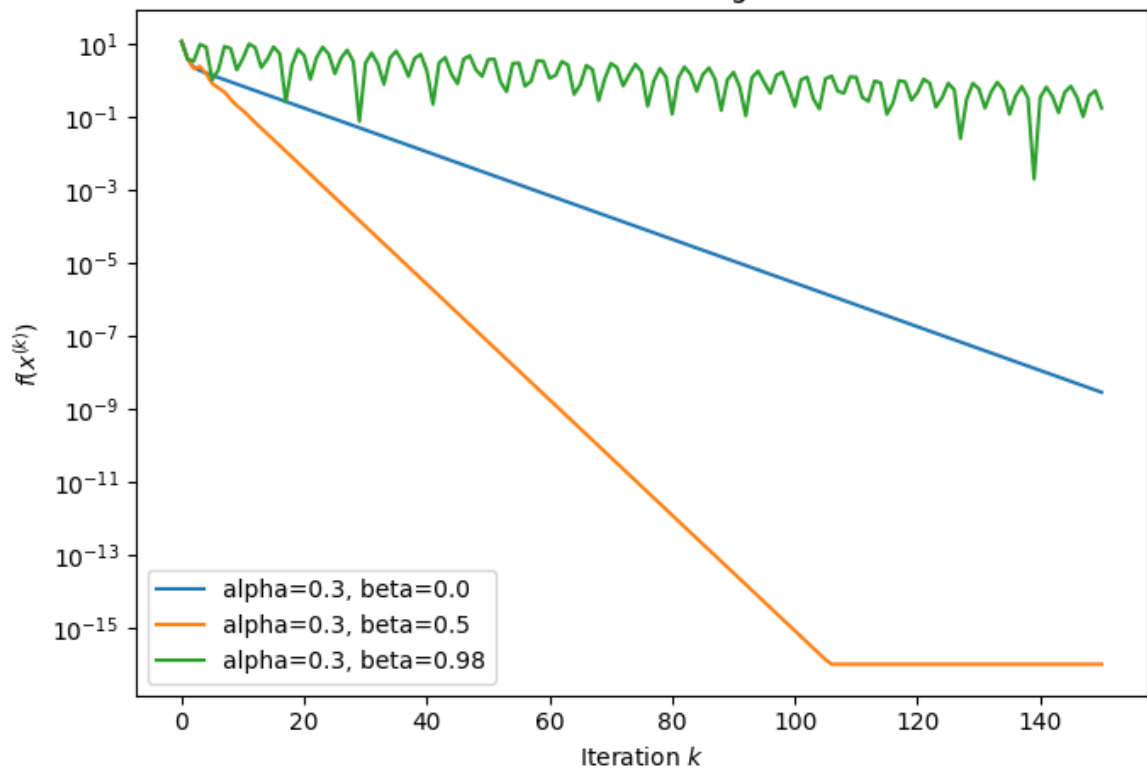
Optimum:

$$x^* = (-1, -1), \quad f_1(x^*) = 0$$

Function 1: Gradient Descent with Momentum



Function 1: Convergence



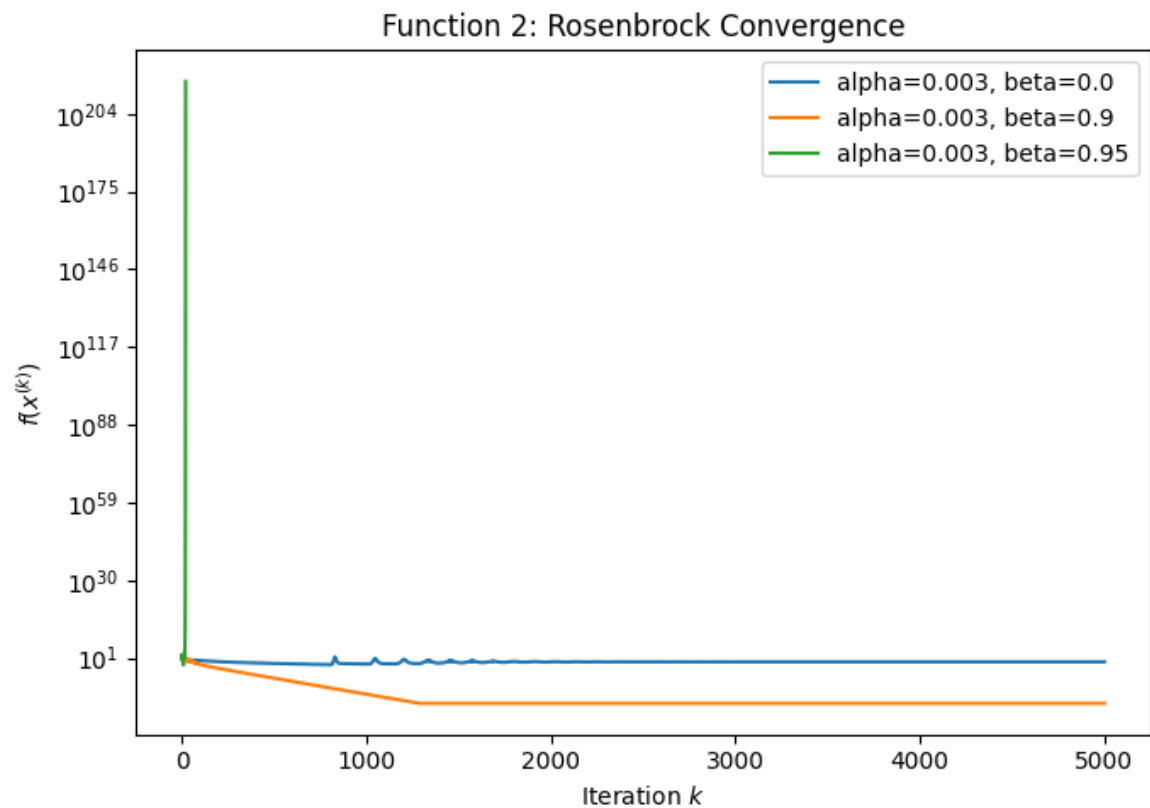
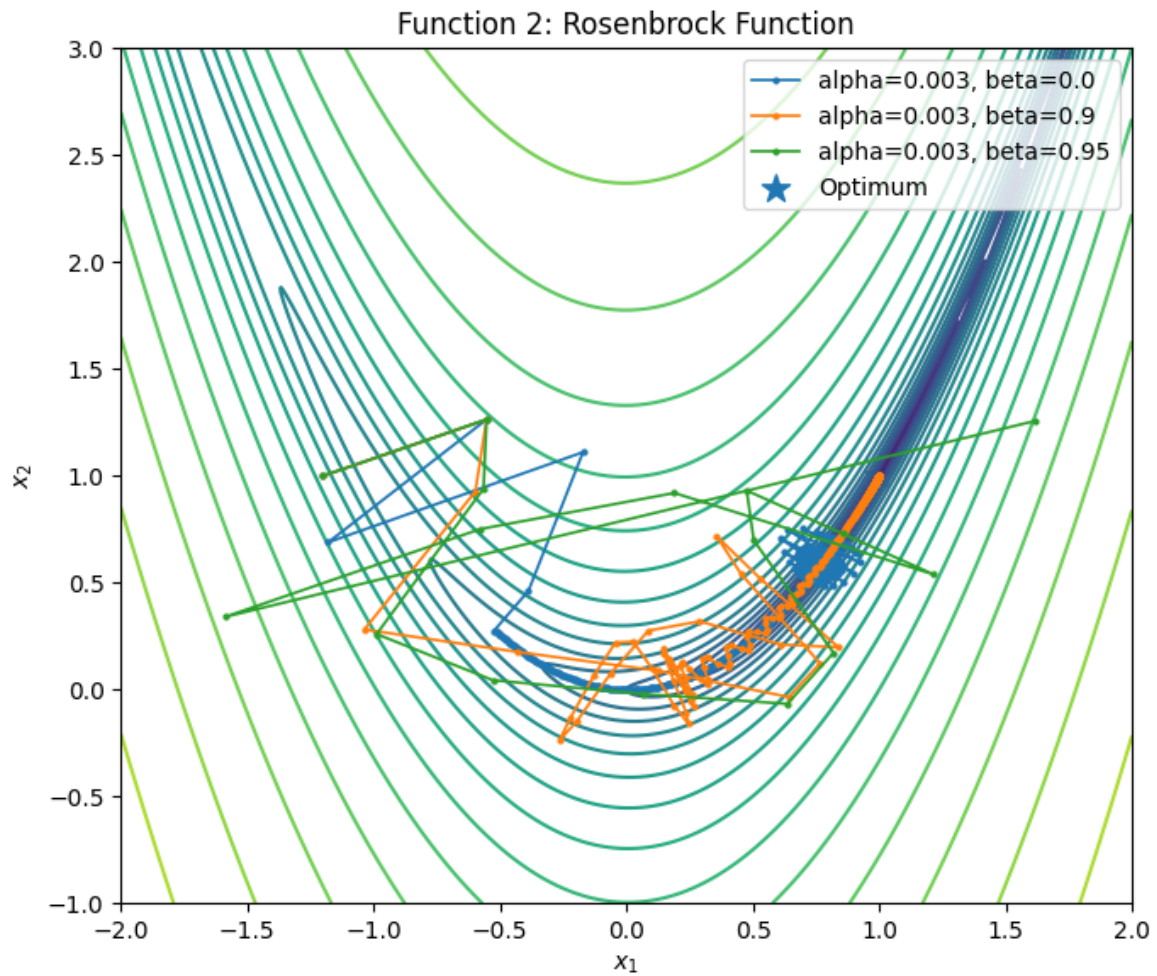
Function 2: Rosenbrock function

$$f_2(x) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$

$$\nabla f_2(x) = [2(x_1 - 1) - 400x_1(x_2 - x_1^2) \quad 200(x_2 - x_1^2)]$$

Optimum:

$$x^* = (1, 1), \quad f_2(x^*) = 0$$



Function 3: own ill-conditioned quadratic function

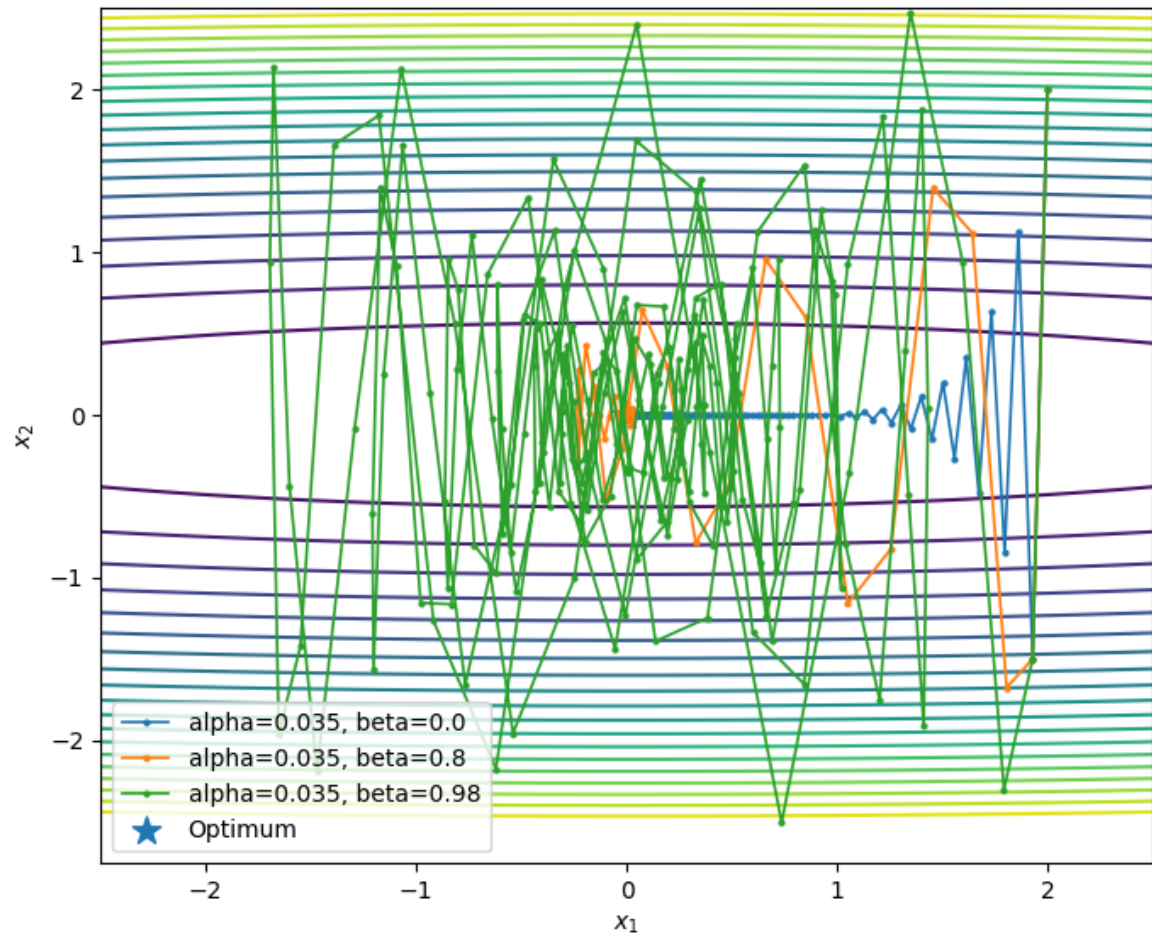
$$f_3(x) = \frac{1}{2}x_1^2 + 25x_2^2$$

$$\nabla f_3(x) = [x_1 \ 50x_2]$$

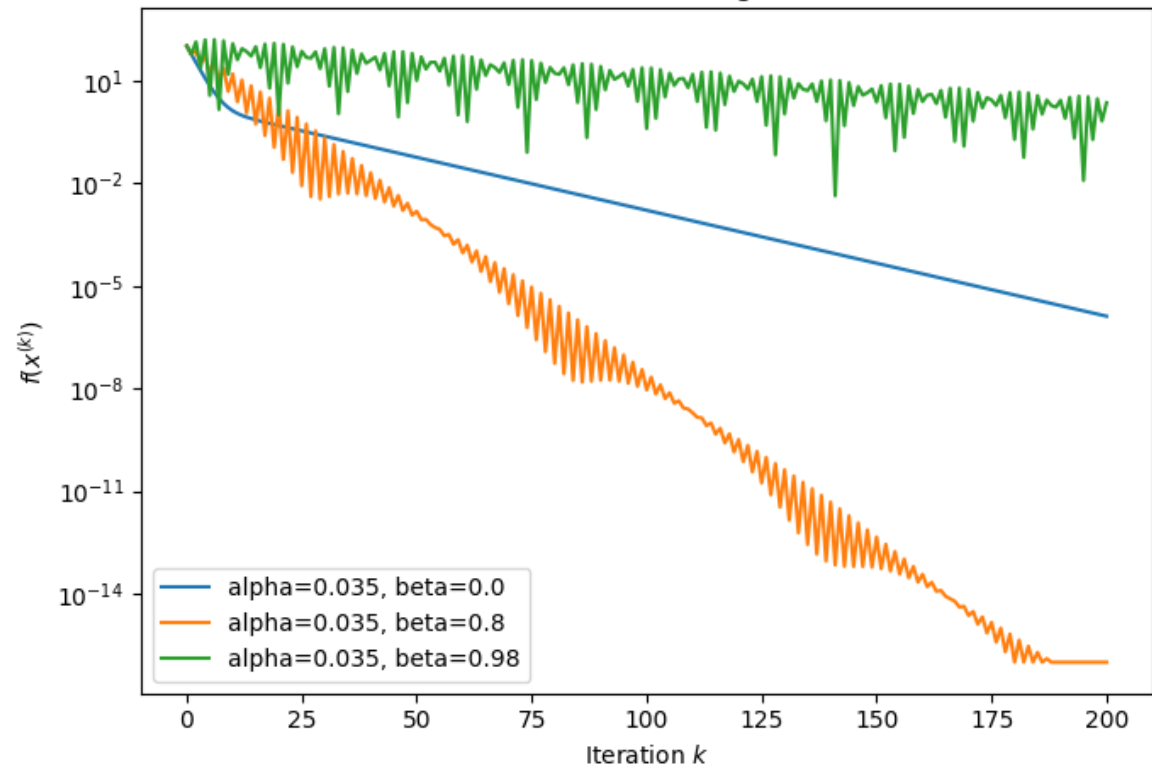
Optimum:

$$x^* = (0, 0), \quad f_3(x^*) = 0$$

Function 3: Own Ill-Conditioned Quadratic



Function 3: Convergence



3. Experimental conclusions

	function	alpha	...	best_f	first_iter_f_less_than_1e-6
0	f1	0.300	...	2.849250e-09	108.0
1	f1	0.300	...	9.979929e-24	43.0
2	f1	0.300	...	2.002808e-03	NaN
3	f2_Rosenbrock	0.003	...	2.781816e-02	NaN
4	f2_Rosenbrock	0.003	...	4.449669e-30	450.0
5	f2_Rosenbrock	0.003	...	NaN	NaN
6	f3_own_function	0.035	...	1.294063e-06	NaN
7	f3_own_function	0.035	...	2.513558e-19	74.0
8	f3_own_function	0.035	...	4.270763e-03	NaN

Function	Setting	Result
f_1	$\alpha = 0.30, \beta = 0$	Reaches $f < 10^{-6}$ after about 108 iterations
f_1	$\alpha = 0.30, \beta = 0.50$	Reaches $f < 10^{-6}$ after about 43 iterations
f_1	$\alpha = 0.30, \beta = 0.98$	Strong oscillation; after 150 iterations, $f \approx 0.178$
f_2	$\alpha = 0.003, \beta = 0$	After 5000 iterations, $f \approx 0.327$
f_2	$\alpha = 0.003, \beta = 0.90$	Reaches $f < 10^{-6}$ after about 450 iterations
f_2	$\alpha = 0.003, \beta = 0.95$	Diverges, showing instability
f_3	$\alpha = 0.035, \beta = 0$	After 200 iterations, $f \approx 1.29 \times 10^{-6}$
f_3	$\alpha = 0.035, \beta = 0.80$	Reaches $f < 10^{-6}$ after about 74 iterations
f_3	$\alpha = 0.035, \beta = 0.98$	Oscillates; after 200 iterations, $f \approx 2.24$

Conclusion: A suitable momentum coefficient accelerates convergence by carrying useful search direction information from previous iterations. This is especially helpful in narrow valleys such as the Rosenbrock function. However, when β is too large, the algorithm has too much inertia, overshoots the minimizer, oscillates, or even diverges.